
STAGE 1
Input Tensor
Raw Trajectory
x in R^24
Process + Patch
STAGE 2
Hygiene Split
Fit Scaler on
Normal-Only Train
Zero Leakage
STAGE 3
Core Model
Bottleneck Autoencoder
24 -> 16 -> 6 -> 16 -> 24
LayerNorm + ReLU
STAGE 4
Scoring s(x)
Reconstruction MSE
s(x) = (1/D) sum |x - x_hat|^2
Scalar Anomaly Score
STAGE 5
Threshold tau
Val-Tuned Cutoff
Cost-Sensitive F2
tau = 0.118
STAGE 6
Operational Alert
Binary Decision
y_hat = 1[s(x) >= tau]
CI/CD Quarantine